A Ridiculously Short Introduction to Derivatives

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Purpose

We want to find the line that best fits a scatterplot. In this context, "best" means the line that makes the total squared error as small as possible. Finding the smallest value of a function requires one main idea from calculus: the derivative.

This handout introduces the four derivative rules used in the least-squares derivation.

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1. What a derivative is

The **derivative** of a function is another function that gives the slope of the original function at each point.

If $f(x)$ is a function, its derivative can be written as $f'(x)$ or $\frac{df}{dx}$. These are two notations for the same quantity.

At a local minimum, the graph is flat, so its slope is zero. Therefore, one way to find an input that minimizes f is to solve $f'(x) = 0$.

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2. The power rule

$$\frac{d}{dx}\, x^n = n\, x^{n-1}$$

To apply the power rule, multiply by the exponent and then reduce the exponent by one.

$f(x)$	$f'(x)$
x^2	$2x$
x^3	$3x^2$
x	1
7 (a constant)	0

A constant has a horizontal graph, so its slope is 0 everywhere.

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3. Constants out front, and sums

$$\frac{d}{dx}\big[c \cdot f(x)\big] = c \cdot f'(x) \qquad\qquad \frac{d}{dx}\big[f(x) + g(x)\big] = f'(x) + g'(x)$$

A constant factor remains in front when differentiating. A sum can be differentiated term by term. The sum rule also applies to longer sums, including those written with $\sum_{i=1}^{n}$.

Example. $f(x) = 3x^2 - 7x + 4 \implies f'(x) = 6x - 7$.

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4. The chain rule

The least-squares derivation frequently involves expressions in which another expression is raised to a power. For these functions, we use the chain rule:

$$\frac{d}{dx}\big[g(x)\big]^n = n\big[g(x)\big]^{n-1} \cdot g'(x)$$

First apply the power rule to the outer expression, and then multiply by the derivative of the inner expression. The second factor, $g'(x)$, is an important part of the rule.

Examples.

- $\frac{d}{dx}(3x+2)^5 = 5(3x+2)^4 \cdot 3 = 15(3x+2)^4$
- $\frac{d}{dx}(7-2x)^2 = 2(7-2x) \cdot (-2) = -4(7-2x)$

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5. Using derivatives to find a minimum

Example. Minimize $f(x) = (x-4)^2 + 1$.

$$f'(x) = 2(x-4) \cdot 1 = 2(x-4) \qquad \implies \qquad 2(x-4) = 0 \qquad \implies \qquad x = 4$$

Solving $f'(x) = 0$ identifies a point where the graph is flat. Such a point could be a minimum, a maximum, or neither. However, the functions used in this unit are sums of squares with upward-opening graphs, so their flat points are minima. For these functions, a second-derivative check is not necessary.

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6. Partial derivatives

The best-fit line has two unknowns: the slope m and the intercept b. Its error function therefore has two inputs, $f(m, b)$, and we need to determine how the function changes with respect to each input.

To take the partial derivative with respect to m, treat b as a constant. Similarly, when taking the partial derivative with respect to b, treat m as a constant.

Partial derivatives use the symbol ∂ rather than d to indicate that the function has other variables that are being held constant:

$$\frac{\partial f}{\partial m} \quad \text{and} \quad \frac{\partial f}{\partial b}$$

Example. $f(m, b) = (7 - 3m - b)^2$

Apply the chain rule in both cases. The derivative of the inner expression depends on which variable we are differentiating with respect to.

- With respect to m, treat b as a constant. The derivative of $7 - 3m - b$ is -3.

$$\frac{\partial f}{\partial m} = 2(7 - 3m - b) \cdot (-3) = -6(7 - 3m - b)$$

- With respect to b, treat m as a constant. The derivative of the inner expression is -1.

$$\frac{\partial f}{\partial b} = 2(7 - 3m - b) \cdot (-1) = -2(7 - 3m - b)$$

To minimize a function of two variables, set both partial derivatives equal to zero and solve the resulting system of equations.

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7. A least-squares example

Consider the two data points $(1, 2)$ and $(3, 5)$. The residual sum of squares for a line $y = mx + b$ is

$$\mathrm{RSS}(m, b) = (2 - m - b)^2 + (5 - 3m - b)^2$$

Use the sum rule, followed by the chain rule on each term, to find both partial derivatives:

$$\frac{\partial \mathrm{RSS}}{\partial m} = -2(2 - m - b) - 6(5 - 3m - b) \qquad \frac{\partial \mathrm{RSS}}{\partial b} = -2(2 - m - b) - 2(5 - 3m - b)$$

Set both derivatives equal to zero and simplify by dividing each equation by -2:

$$17 - 10m - 4b = 0 \qquad 7 - 4m - 2b = 0$$

Solving the system gives $m = 1.5$ and $b = 0.5$. This agrees with the slope and intercept of the line through $(1, 2)$ and $(3, 5)$. With only two data points, the best-fit line passes through both points, so the residual sum of squares is zero.

For n data points, the same rules apply, with the terms collected in a sum:

$$\frac{\partial}{\partial m} \sum_{i=1}^{n} \big(y_i - mx_i - b\big)^2 = \sum_{i=1}^{n} 2\big(y_i - mx_i - b\big) \cdot (-x_i)$$

This expression uses the chain rule; the derivative of the inner expression with respect to m is $-x_i$.

Practice

1. $f(x) = 5x^3 - 2x + 11$. Find $f'(x)$.
2. $f(x) = (4x - 9)^2$. Find $f'(x)$.
3. Minimize $f(x) = (x - 6)^2 + (x - 2)^2$.
4. $f(m, b) = (10 - 2m - b)^2$. Find $\partial f/\partial m$ and $\partial f/\partial b$.
5. $f(m, b) = 4m^2 b + b^3 - m$. Find both partial derivatives.

Answers

1. $15x^2 - 2$
2. $2(4x - 9) \cdot 4 = 8(4x - 9) = 32x - 72$
3. $f'(x) = 2(x - 6) + 2(x - 2) = 4x - 16$; setting this equal to zero gives $x = 4$, the mean of 6 and 2.
4. $\partial f/\partial m = -4(10 - 2m - b)$; $\partial f/\partial b = -2(10 - 2m - b)$
5. $\partial f/\partial m = 8mb - 1$; $\partial f/\partial b = 4m^2 + 3b^2$